

BIB Week 1 — What is left for a bioengineer to do

- Biyomühendise kalan iş: 2026'da biyoinformatik
- Ege Üniversitesi, Biyomühendislik Bölümü — 4. sınıf, zorunlu
- Yasin Kaymaz, PhD
- Bring your laptop out now. You will need it in the second half.

Ege Üniversitesi · Biyomühendislik Bölümü

BIB · Bioinformatics for Bioengineering · Week 1

What is left for a bioengineer to do

Biyomühendise kalan iş: 2026'da biyoinformatik

4. sınıf, zorunlu · 2026–27 · Yasin Kaymaz, PhD

EN / TR

Ad of the Week #1

- One real job advert, every week, five minutes, no exceptions
- This week: read the requirements list out loud
- Tick what you already have on your Skills Ledger (Beceri Defteri)
- This is the entire careers thread. There is no careers lecture.

The Skills Ledger (Beceri Defteri)

- One A4 page. Keep it. Bring it to Week 14.
- Left column: the skill. Right column: date first done, and where the evidence is.
- You tick a line only when you can point at a notebook that proves it
- Ungraded until a two-minute check in Week 14
- It is not a to-do list. It is a receipt for what you already did.

SPECIFICATION · CONTROLS · ADJUDICATION · ACCOUNTABILITY

Skills Ledger (Beceri Defteri) · Name: _____ · Student no: _____ · Seed: _____

#	Skill ('I can...' / 'I have...') — tick only with evidence	First done	Evidence (notebook)
1	State the BED coordinate convention — and prove it		
2	Turn a BED interval into GTF coordinates, length right		
3	Cite a number with resource, release and access date		
4	Fact-check a chat model's claim against GENCODE		
5	Resolve a DOI a model gave me — and catch a fake one		
6	Read a Python traceback bottom-up, name the failing line		
7	Find a semantic defect in code I did not write		
8	Score a short alignment by hand before running the tool		
9	Say why two alignment tools disagree, and pick one		
10	Design replicates vs depth under a fixed read budget		
11	Say what Phred Q30 means as a number		
12	Write my expected answer down before running anything		
13	Build a control whose answer I already know		
14	Read a p-value histogram and say what its shape means		
15	Normalise counts in the right order and explain why		
16	Audit an AI-written analysis plan claim by claim		
17	Shuffle the labels to control a classifier's accuracy		
18	Spot leakage in a train/test split		
19	Write an AI-use disclosure a reviewer could act on		
20	Defend one of my own numbers in a five-minute viva		

bring to Week 14

Skills Ledger (Beceri Defteri) — tick a line only when a notebook proves it. Bring to Week 14.

How many genes does a human have?

- Say a number out loud. Any number.
- I will take three answers and write them on the board.
- Then we will go and look.

Live: the GENCODE human statistics page

- <https://www.gencodegenes.org/human/stats.html>
- Read the release banner FIRST, before any number
- Then: protein-coding genes / total gene entries / transcripts

GENCODE 50, June 2026 — the number set

- 19,442 protein-coding genes
- 78,733 total gene entries
- 644,292 transcripts; 278,455 of them protein-coding
- 35,885 lncRNA genes · 7,608 small ncRNA genes · 14,702 pseudogenes
- Release: GENCODE 50 on GRCh38.p14.
- Source:
<https://www.gencodegenes.org/human/stats.html>

GENCODE Release 50 (GRCh38.p14) · June 2026

Category	Count
Protein-coding genes (protein kodlayan gen)	19,442
Total gene entries	78,733
lncRNA genes	35,885
Small ncRNA genes	7,608
Pseudogenes	14,702
Transcripts, all (transkript)	644,292
Transcripts, protein-coding	278,455

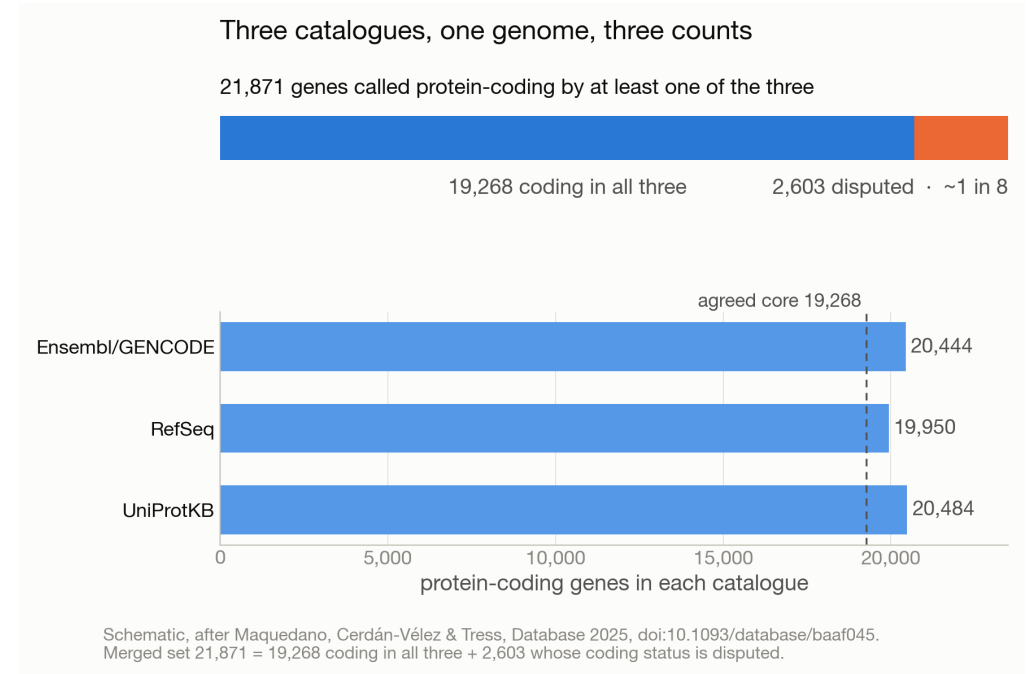
Source: <https://www.gencodegenes.org/human/stats.html>

Accessed: 2026-09-14 · next June this table will be wrong

*GENCODE Release 50 (GRCh38.p14) · <https://www.gencodegenes.org/human/stats.html>
· accessed 2026-09-14*

The catalogues do not agree

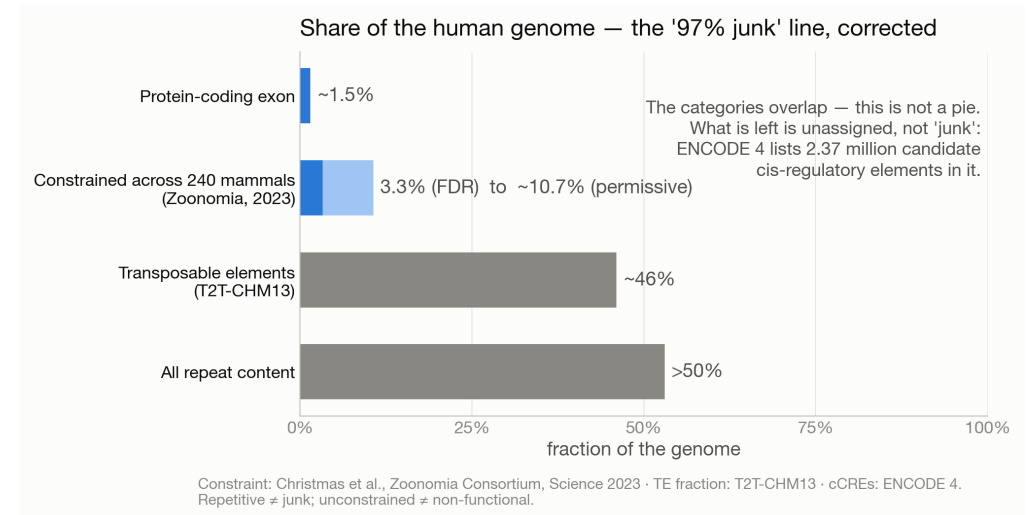
- Ensembl/GENCODE, RefSeq and UniProtKB annotate 21,871 coding genes between them
- Union of all three: 22,210
- They disagree on the coding status of 2,603 genes — roughly one in eight
- So the defensible sentence is: "about 19,000–20,000, depending on which catalogue, and they still argue about ~12%"
- Maquedano, Cerdán-Vélez & Tress, Database 2025, doi:10.1093/database/baaf045



*Schematic, after Maquedano, Cerdán-Vélez & Tress, Database 2025,
doi:10.1093/database/baaf045*

And while we are here: delete "97% junk DNA"

- ~1.5% of the genome is protein-coding exon
- ~3.3% of bases constrained at FDR; ~10.7% on a permissive estimate (Zoonomia, Science 2023)
- Transposable elements alone: ~46% of T2T-CHM13; total repeat content >50%
- ENCODE's "80% biochemical activity" meant reproducible biochemical signature, and was walked back
- ENCODE 4 registry: 2.37 million candidate cis-regulatory elements in human
- "Junk" is not a defensible category.
- "Unconstrained" is.



*Share of the human genome — overlapping categories, not a pie.
Zoonomia (Science 2023), T2T-CHM13, ENCODE 4.*

The rule that runs all semester

- Every number carries three things:
 - the resource (which database)
 - the release (which version of it)
 - the access date (when you looked)
- A number without those three is a rumour
- This is the G and the S of the GENOMES card, which you get next week
- It applies to numbers from a paper, from a database, and from a model

Every number carries three things

19,442 protein-coding genes

RESOURCE Gencode · <https://www.encodegenes.org/human/stats.html>

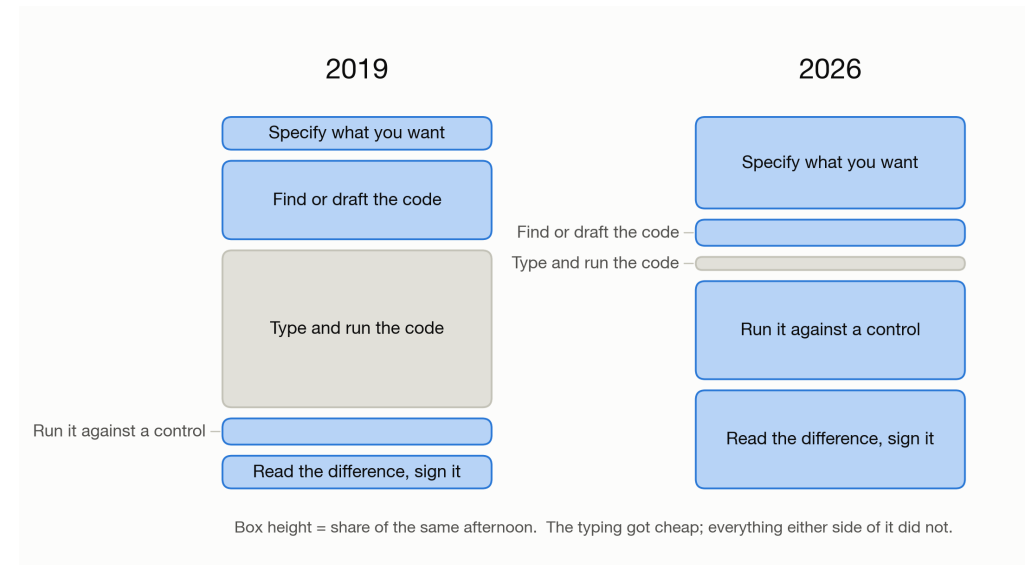
RELEASE Release 50 (GRCh38.p14)

ACCESS DATE 2026-09-14

A number without these three is a rumour.

The 2026 work loop

- 2019: read the docs, search, adapt a snippet, run it, stare at the output
- 2026:
 - write a short specification (§artname)
 - let a model draft the code
 - run it against a control (kontrol) whose answer you already know
 - read the difference, and take responsibility
- The typing got cheap. Everything either side of the typing did not.



Box height = share of the same afternoon.

The four irreducibly human jobs

- Specification — saying precisely what you want, including what "correct" means
- Controls — building the thing whose answer you already know
- Adjudication — deciding which of two plausible outputs is right
- Accountability — your name is on it (hesap verebilirlik)
- Learn these four words. They are the spine of this course.

Specification

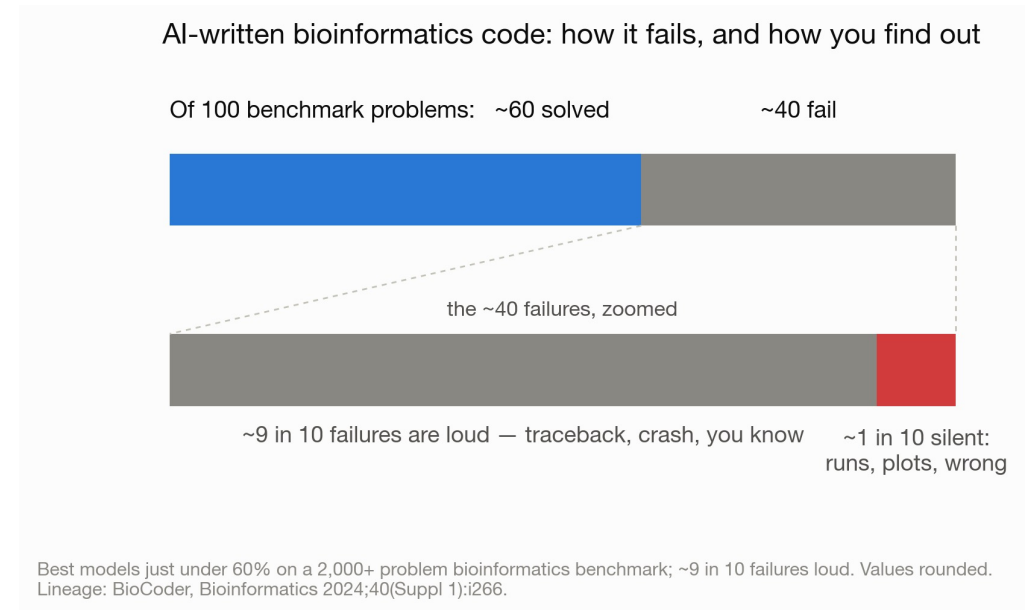
Controls

Adjudication

Accountability

One honest number about AI-written code

- On a bioinformatics coding benchmark of 2,000+ problems, the best models top out just under 60%
- Roughly 9 in 10 of the failures are loud — syntax errors, tracebacks, crashes
- Which means about 1 in 10 are silent
- Silent is the dangerous kind. Loud failures cost you an afternoon. Silent ones reach a paper.
- Benchmark lineage: BioCoder, Bioinformatics 2024, 40(Suppl 1):i266 — https://academic.oup.com/bioinformatics/article/40/Supplement_1/i266/7700865



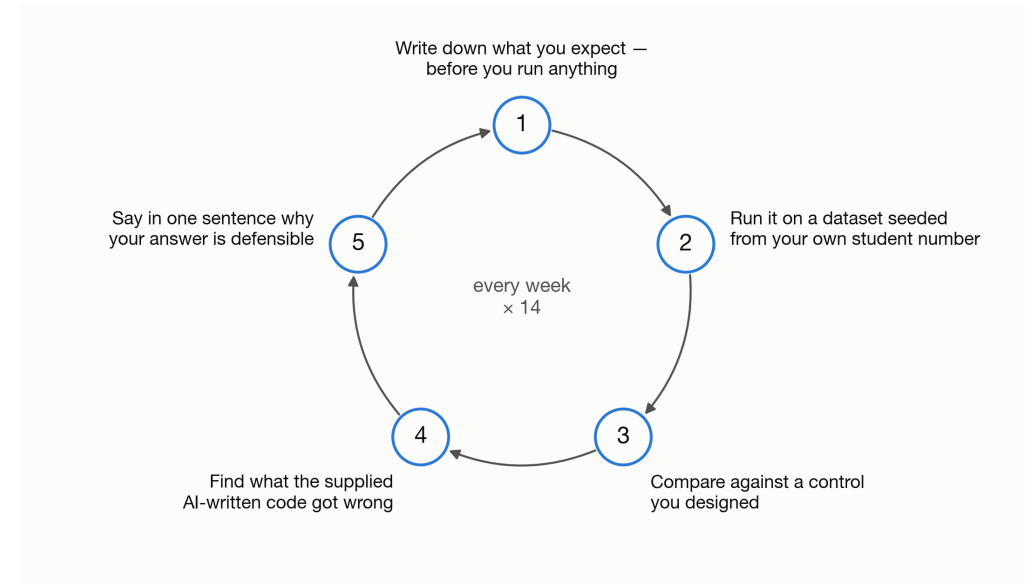
Lineage: BioCoder, Bioinformatics 2024;40(Suppl 1):i266. Values rounded.

The course contract

- Five things you do, every week, for fourteen weeks
- One assessment map
- One AI policy, in four colours, with a reason attached to each
- One honest workload number

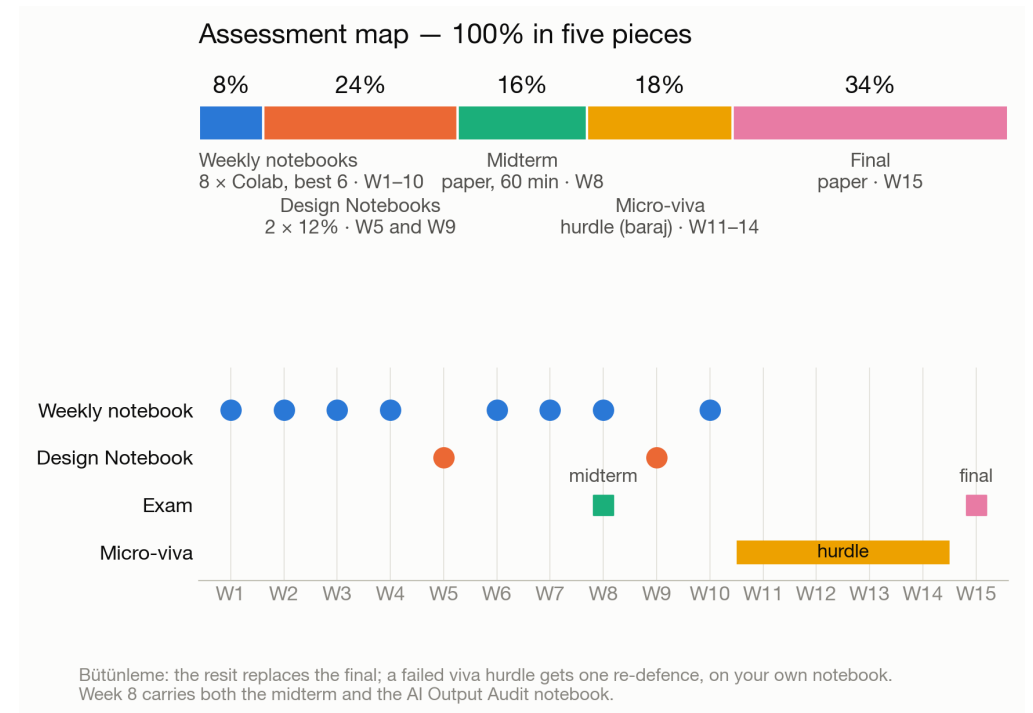
The five things you do every week

- Write down what you expect, before you run anything
- Run something on a dataset seeded from your own student number
- Compare it against a control you designed
- Find what the supplied AI-written code got wrong
- Say in one sentence why your answer is defensible
- Every lecture exists only to give you the judgement that week's loop needs
- Nothing is taught here that no loop consumes



Assessment map

- Weekly Colab notebooks — 8% (eight notebooks, best six count)
- Design Notebooks 1 and 2 (Weeks 5 and 9) — 24% total, 12% each
- Midterm, Week 8, on paper, 60 minutes — 16%
- Final, Week 15, on paper — 34%
- Individual micro-viva on your own notebook, Weeks 11–14 — 18%, and it is a hurdle (baraj)
- Bütünleme: the resit replaces the final; a failed viva hurdle gets one re-defence, on your own notebook



The AI policy: four colours, and why

- RED — midterm, final, bütünleme, micro-viva.
Supervised room, paper or face-to-face.
- YELLOW — the eight weekly notebooks. AI explains; it does not answer. Your seeded data never goes into it.
- BLUE — the two Design Notebooks and the Week 8 AI Output Audit. AI use is required, and you submit the record.
- GREEN — revision, translation of your own writing, the Skills Ledger, optional material.
- Every submitted notebook has an ### AI DISCLOSURE cell. No disclosure, no receipt code.
- Menu, and the duty to justify it: MIT Ad Hoc Committee on AI Use, Rec. 3.1.8, Aug 2026

The AI policy: four colours, and why

	Tier	Which components	Why
	RED	Midterm · Final · Bütünleme Micro-viva	Prohibition is only real where the room is supervised — paper or face-to-face.
	YELLOW	The eight weekly notebooks	AI explains errors and concepts. It does not produce your answer; your seeded data never goes in.
	BLUE	Design Notebooks 1 & 2 Week 8 AI Output Audit	AI use is required and the interaction record is submitted. 'It got nothing wrong' is marked down.
	GREEN	Revision · translating your own writing · Skills Ledger · extras	Not graded on authorship; restricting it would only disadvantage students who need language support.

Every submitted notebook has an ### AI DISCLOSURE cell. No disclosure, no receipt code.
This course does not use AI-detection software.
Menu and the duty to justify it: MIT Ad Hoc Committee on AI Use in Teaching, Learning and Research Training, Rec. 3.1.8 and Appendix B, August 2026.

*Menu and the duty to justify it: MIT Ad Hoc Committee on AI Use,
Rec. 3.1.8 and Appendix B, Aug 2026*

The honest workload number

- 3 ECTS. Roughly 78 total student hours for the semester.
- About 39 of those are contact hours you are sitting in.
- That leaves ~2 hours per week out of class. That is the budget I designed to.
- If you are spending five hours a week on this, something is wrong — tell me, do not suffer
- There is no team project and no 4-week pipeline build this year. Both were deleted.

Live onboarding — everybody, now, together

- Five steps. I will read each one aloud and wait.
- We do this here, in the lecture hall, while everyone is present
- Not in the application hour. If eleven of you cannot log in, I need to know now.
- When you finish step 5: green sticky note on your laptop lid, then help the person next to you

- 1 Open the link
EgeDers › BIB 2026 › Week 1 › W01 — hello_seed (mirror: the GitHub badge)
- 2 File › Save a copy in Drive
the tab title must now say Copy of W01_hello_seed.ipynb
- 3 Run the setup cell — Shift+Enter
expect: bib_colab ready — course BIB2026 and a green tick
- 4 A.whoami(name, student number, section)
refused? hand up — it is a typo or a late enrolment, fixed here
- 5 Write your seed on paper
then a green sticky note on the lid, and help the person next to you

Step 1 — Open the link

- Go to EgeDers → BIB 2026 → Week 1
- Click the single link named W01 — hello_seed
- It opens Colab in your browser. Sign in with your Google account if asked.
- Mirror, if EgeDers is slow: the GitHub badge on the same EgeDers page

Step 2 — File → Save a copy in Drive

- File → Save a copy in Drive
- Do it NOW, before you run anything
- The tab title should change to Copy of W01_hello_seed.ipynb
- If it does not say "Copy of", you are working in my file and your work will not save

Step 3 — Run the setup cell

- Click the first cell. Press Shift+Enter.
- It downloads one file, `bib_colab.py`, about 6 KB, from EgeDers
- No pip install. Nothing to resolve. Nothing to break.
- You should see: `bib_colab ready — course BIB2026`
- A green tick appears to the left of the cell

Step 4 — A.whoami(...)

- Fill in your name, your student number, and your section (EN or TR)
- Run the cell
- It checks your number against the class roster and refuses an unknown number
- If it refuses you, raise your hand. We fix it here, in ten seconds.
- Better here than at 23:40 on Sunday night

roster OK — carry on

```
[2] A.whoami(  
    name      = "AD SOYAD",  
    student_no = "05220000000",  
    section    = "TR",  
)
```

output

welcome, AD SOYAD · section TR · roster OK

refused — the cell is working correctly

```
[2] A.whoami(  
    name      = "AD SOYAD",  
    student_no = "0522000000",  
    section    = "TR",  
)
```

output

student number not found in roster –
check the digits, then ask the instructor

one digit short

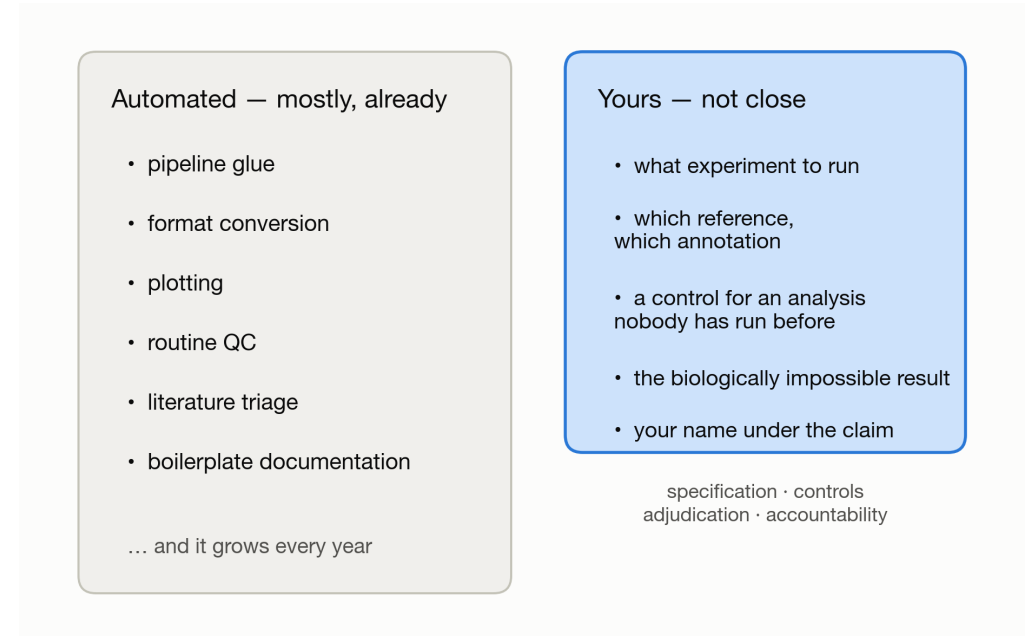
Mock-up of the expected output.

Step 5 — Your seed

- The cell prints your seed (tohum) and your dataset ID
- Write it down on paper. Now. On the Skills Ledger, in the corner.
- This number generates your dataset for the whole semester
- It is the reason no two of you have the same answer
- Green sticky note on your lid. Then look left and right and help someone.

What is being automated, and what is not

- Automated, mostly, already: pipeline glue, format conversion, plotting, routine QC, literature triage
- Not automated, and not close:
 - deciding what experiment to run
 - choosing the reference and the annotation
 - building a control for an analysis nobody has run before
 - recognising the biologically impossible result
 - signing your name under a claim
- The four words again: specification · controls · adjudication · accountability



Friday's lab, and what is due

- Lab (application hour): get your gene, fact-check a chat model against GENCODE 50, disclose, submit
- Bring: laptop, your seed, the Skills Ledger
- A.finish() happens at minute 30 of the lab. Everything after that line is optional and at home.
- Due Sunday 23:59 — download the .ipynb, upload to the Week 1 Form with your receipt code
- Missed the onboarding today? Sit in the front row on Friday.

Every notebook this semester opens with this header

Step 0 — Setup

```
[ ] # Fetches the small helper module the course uses (~6 KB). Run once.
    !curl -s0 https://egeders.ege.edu.tr/BIB2026/bib_colab.py
    # mirror, only if EgeDers is unreachable:
    # !curl -s0 https://raw.githubusercontent.com/BMGLab/BIB/main/bib_colab.py

import numpy as np
from scipy import stats
import matplotlib.pyplot as plt
import bib_colab as bib

A = bib.start("W01")
```

Step 1 — Identify yourself

```
[ ] A.whoami(
    name      = "YOUR NAME HERE",
    student_no = "05220000000",   # <- your Ege student number, digits only
    section    = "EN",            # <- "EN" or "TR"
)
```

Step 2 — Your dataset

```
[ ] rng, params = A.dataset()
```

Same header in every week (Step 0–2). Only the assignment code in `bib.start(...)` changes from week to week.

Same header in every week (Step 0–2). Only the assignment code in `bib.start(...)` changes from week to week.

Sources on this slide set

- GENCODE Release 50 statistics (GRCh38.p14) — <https://www.encodegenes.org/human/stats.html>
- Maquedano, Cerdán-Vélez & Tress, "The state of the human coding gene catalogues", Database 2025 — doi:10.1093/database/baaf045
- Christmas et al., Zoonomia Consortium, Science 2023 — <https://www.science.org/doi/10.1126/science.abn3943>
- BioCoder benchmark lineage, Bioinformatics 2024;40(Suppl 1):i266 — https://academic.oup.com/bioinformatics/article/40/Supplement_1/i266/7700865
- Teufel & Liberles, "Teaching bioinformatics with generative AI", JMBE 2026 — doi:10.1128/jmbe.00331-25
- Liang et al., "GPT detectors are biased against non-native English writers", Patterns 2023 — arXiv:2304.02819
- MIT Ad Hoc Committee on AI Use in Teaching, Learning and Research Training, August 2026 — Rec. 3.1.8 and Appendix B